CLASS 12 SCIENCE

PCM • COMPLETE STUDY MATERIAL

BIHAR SCHOOL EXAMINATION BOARD (BSEB)
Academic session 2026–27 • English Medium

PHYSICS
Concept summaries • Key formulae • Numerical approach

CHEMISTRY
Physical • Inorganic • Organic chemistry

MATHEMATICS
Core concepts • Methods • Essential identities

A structured revision companion for board-exam preparation

Important: This is a learning and revision guide, not a replacement for the latest official BSEB syllabus, prescribed textbook, school instructions or practical manual. Chapter inclusion and depth can change; confirm the final examinable topics with your school and the board's current syllabus.

How to use this book

- Study one chapter at a time and write definitions, laws and derivations in your own words.
- For Physics, draw a labelled diagram, write the governing law, substitute values with SI units and check the final unit.
- For Chemistry, practise balanced equations, conditions, named reactions, structures and numerical problems.
- For Mathematics, show every step, state the formula used and verify the result wherever possible.
- Keep a separate error notebook. Re-solve incorrect questions after one day and again after one week.
- Use the latest BSEB model paper and previous-year papers for question style and time management.

Contents

Part	Coverage
I. Physics	Electrostatics, current electricity, magnetism, EMI/AC, EM waves, optics, modern physics, semiconductors
II. Chemistry	Solid state, solutions, electrochemistry, kinetics, surface chemistry, metallurgy, p/d/f block, coordination, organic chemistry, biomolecules and polymers
III. Mathematics	Functions, matrices, determinants, calculus, differential equations, vectors, 3D geometry, linear programming and probability
IV. Exam practice	Study plan, answer-writing checklist, sample questions and practical preparation

PART I — PHYSICS

How to study: Read the concept points, learn the boxed relations, solve textbook examples, then attempt the practice questions at the end of this subject.

1. Electric Charges and Fields

- Electric charge is conserved, quantised (q = ne) and additive. Like charges repel; unlike charges attract.
- Coulomb's law: force between point charges acts along their line of centres and varies inversely as the square of separation. In a medium, replace ε_0 by ε.
- Electric field is force per unit positive test charge. Field lines start on positive charges and end on negative charges; they never intersect.
- An electric dipole has equal and opposite charges separated by a small distance. Dipole moment p = q × separation, directed from − to +.
- Electric flux measures the field passing through a surface. Gauss's law relates net flux to enclosed charge; it is especially useful for spherical, cylindrical and planar symmetry.

Key formulae / facts

$F = (1/4\pi\varepsilon_0)\, q_1 q_2 / r^2$

$E = F/q_0$; point charge: $E = (1/4\pi\varepsilon_0) q/r^2$

$\Phi = E \cdot A = EA\cos\theta$

$\oint E \cdot dA = Q_{enclosed}/\varepsilon_0$

Dipole torque: $\tau = pE\sin\theta$

2. Electrostatic Potential and Capacitance

- Potential is work done per unit positive test charge in bringing it from infinity. Potential is scalar; electric field is vector.
- For a point charge V = kq/r. Potential difference is related to field by E = −dV/dr in one dimension.
- A capacitor stores charge and energy. Capacitance depends on geometry and dielectric, not on Q or V separately.
- For capacitors in parallel, potential difference is common; in series, charge magnitude is common. A dielectric increases capacitance by its dielectric constant when it fully fills the gap.

Key formulae / facts

$V = W/q$; $V = kq/r$

$C = Q/V$

Parallel plate: $C = \varepsilon_0 A/d$ (×K with dielectric)

Parallel: $C_{eq} = \Sigma C$; Series: $1/C_{eq} = \Sigma(1/C)$

$U = \frac{1}{2}CV^2 = Q^2/(2C) = \frac{1}{2}QV$

3. Current Electricity

- Electric current is the rate of flow of charge. Conventional current is opposite to electron drift.
- Ohm's law holds for ohmic conductors at constant physical conditions. Resistance depends on material, length and area.
- Resistivity is a material property; conductivity is its reciprocal. Temperature generally increases metallic resistance.
- Kirchhoff's junction rule expresses charge conservation; loop rule expresses energy conservation.
- Cells in series add emfs; internal resistance must be included. Wheatstone bridge is balanced when the ratio of resistances in its arms is equal.

Key formulae / facts

$I = dQ/dt$; $I = neAv_d$

$V = IR$; $R = \rho L/A$

$P = VI = I^2R = V^2/R$

Series: $R_{eq} = \Sigma R$; Parallel: $1/R_{eq} = \Sigma(1/R)$

Terminal voltage: $V = E - Ir$ (discharging)

4. Moving Charges and Magnetism

- A moving charge experiences magnetic force perpendicular to both velocity and magnetic field. A stationary charge experiences no magnetic magnetic force.
- A current-carrying conductor in a magnetic field experiences a force. The direction is found using the appropriate hand rule.
- A charged particle entering a uniform magnetic field perpendicular to it follows a circular path; magnetic force does no work.
- A current loop behaves like a magnetic dipole. A galvanometer detects small currents and can be converted to an ammeter or voltmeter.

Key formulae / facts

$F = q(v \times B)$; $|F| = qvB\sin\theta$

$F = I(L \times B)$; $|F| = BIL\sin\theta$

$r = mv/(|q|B)$

B at centre of N-turn coil = $\mu_0 NI/(2R)$

$\tau = NIAB\sin\theta$; $\mu = NIA$

5. Magnetism and Matter

- A bar magnet is equivalent to a solenoid in its external field pattern. Magnetic field lines form closed loops.
- Earth behaves approximately like a giant magnet. Magnetic elements include declination, dip and horizontal component.
- Diamagnetic materials are weakly repelled, paramagnetic materials weakly attracted, and ferromagnetic materials strongly attracted.
- In a uniform field, a magnetic dipole experiences torque but no net force.

Key formulae / facts

$\tau = mB\sin\theta$

Potential energy $U = -m \cdot B$

$B = \mu_0(H + M)$

$\chi_m = M/H$; $\mu_r = 1 + \chi_m$

6. Electromagnetic Induction

- A changing magnetic flux through a circuit induces an emf. The induced current opposes the change that produces it (Lenz's law).
- The magnitude of induced emf is the rate of change of magnetic flux linkage. Motional emf occurs when a conductor moves through a magnetic field.
- Self-inductance measures the flux linkage produced per unit current. An inductor opposes changes in current.

Key formulae / facts

$\Phi_B = BA\cos\theta$

$\varepsilon = -N(d\Phi_B/dt)$

Motional emf: $\varepsilon = Blv$ (mutually perpendicular)

$\varepsilon_L = -L(dI/dt)$

Energy in inductor: $U = \frac{1}{2}LI^2$

7. Alternating Current

- An alternating current changes magnitude and direction periodically. For a sinusoid, rms value is peak value divided by √2.
- In a pure resistor, current and voltage are in phase; in an inductor current lags; in a capacitor current leads.
- In a series LCR circuit, impedance depends on resistance and net reactance. At resonance, XL = XC and current is maximum.
- A transformer works on mutual induction and changes AC voltage according to turns ratio; it does not operate on steady DC.

Key formulae / facts

$V_{rms} = V_0/\sqrt{2}$; $I_{rms} = I_0/\sqrt{2}$

$X_L = \omega L$; $X_C = 1/(\omega C)$

$Z = \sqrt{[R^2 + (X_L - X_C)^2]}$

$\tan\varphi = (X_L - X_C)/R$

Resonance: $\omega_0 = 1/\sqrt{(LC)}$; $V_s/V_p = N_s/N_p$

8. Electromagnetic Waves

- A time-varying electric field produces a magnetic field and vice versa. Electromagnetic waves are transverse and need no material medium.
- In vacuum, all electromagnetic waves travel at c. Their spectrum is ordered by wavelength/frequency.
- Radio waves are used in communication; microwaves in radar and cooking; infrared in thermal imaging; visible light in vision; ultraviolet in sterilisation; X-rays in imaging; gamma rays in nuclear processes.

Key formulae / facts

$c = 1/\sqrt{(\mu_0\varepsilon_0)}$

$c = \nu\lambda$

$E_0/B_0 = c$

9. Ray Optics and Optical Instruments

- Reflection obeys angle of incidence = angle of reflection. Refraction follows Snell's law.
- A lens forms images by refraction at two spherical surfaces. Sign convention must be applied consistently.
- Total internal reflection requires light to travel from denser to rarer medium and incidence angle to exceed critical angle.
- A compound microscope and astronomical telescope increase angular size. Their magnifying power depends on focal lengths and adjustment.

Key formulae / facts

$n_1 \sin i = n_2 \sin r$

Mirror: $1/f = 1/v + 1/u$; $m = -v/u$

Lens: $1/f = 1/v - 1/u$; $m = v/u$

Power: $P = 1/f$ (f in metres), unit dioptre

Critical angle: $\sin C = n_2/n_1$ ($n_1 > n_2$)

10. Wave Optics

- Huygens' principle treats each point on a wavefront as a source of secondary wavelets.
- Interference is redistribution of intensity due to superposition of coherent waves. Constructive and destructive conditions depend on path difference.

- In Young's double-slit experiment, fringe width increases with wavelength and screen distance and decreases with slit separation.
- Diffraction is bending/spreading of waves around obstacles or apertures. Polarisation demonstrates the transverse nature of light.

Key formulae / facts

Constructive: Δ = nλ; destructive: Δ = (n + ½)λ

Fringe width: β = λD/d

Single-slit minima: a sinθ = nλ (n = 1,2,...)

11. Dual Nature of Radiation and Matter

- Photoelectric emission supports the particle nature of light. Emission occurs only above a threshold frequency.
- Einstein's photoelectric equation balances photon energy with work function and maximum kinetic energy.
- Matter waves are associated with moving particles; wavelength decreases as momentum increases.

Key formulae / facts

E_photon = hν = hc/λ

K_max = hν − φ = eV₀

λ_de Broglie = h/p

For electron accelerated through V: λ = h/√(2meV)

12. Atoms and Nuclei

- Rutherford scattering established a small, dense, positively charged nucleus. Bohr's model explains hydrogen-like spectra using quantised orbits.
- Nuclear radius varies approximately as A^(1/3). Mass defect gives binding energy; greater binding energy per nucleon generally means greater stability.
- Radioactive decay is spontaneous and follows an exponential law. Fission splits a heavy nucleus; fusion combines light nuclei.

Key formulae / facts

r_n = r₀A^(1/3)

E_n = −13.6 Z²/n² eV (hydrogen-like)

N = N₀e^(−λt)

T_½ = 0.693/λ; mean life = 1/λ

E_bind = Δmc²

13. Semiconductor Electronics

- Conductors, insulators and semiconductors differ by their energy bands. Doping creates n-type (electrons majority) or p-type (holes majority) material.
- A p-n junction conducts strongly in forward bias and weakly in reverse bias. A diode is used for rectification.
- A Zener diode is used as a voltage regulator. Transistors can act as amplifiers and switches; logic gates implement Boolean operations.

Key formulae / facts

I_E = I_B + I_C

α = I_C/I_E; β = I_C/I_B

β = α/(1 − α)

AND: Y = A·B; OR: Y = A + B; NOT: Y = Ā

Physics: numerical-solving checklist

- Write the given quantities and convert them into SI units before substitution.
- Identify the physical law or equation and state any assumptions.
- Draw a diagram or circuit where relevant; mark directions and polarities.
- Keep significant figures sensible and include the correct unit in the final answer.
- For derivations, show the logical steps rather than writing only the final formula.

Physics practice questions

- State Gauss's law and explain one symmetric application.
- Derive the expression for equivalent resistance in series and parallel.
- Explain the principle and working of a moving-coil galvanometer.
- State Faraday's laws and explain Lenz's law using conservation of energy.
- Derive the relation for impedance in a series LCR circuit and state the resonance condition.
- Explain Young's double-slit experiment and obtain the fringe-width expression.
- Explain the photoelectric effect and write Einstein's equation.
- Distinguish between intrinsic and extrinsic semiconductors; explain a p-n junction diode.

Practical preparation

Revise the aim, apparatus, labelled circuit/optical diagram, formula, observation table, graph, calculation, result and precautions for every experiment prescribed by your school. Practise least count, zero error and significant figures.

PART II — CHEMISTRY

How to study: Read the concept points, learn the boxed relations, solve textbook examples, then attempt the practice questions at the end of this subject.

1. The Solid State

- Crystalline solids have long-range order; amorphous solids lack it. Unit cells describe repeating crystal structure.
- For cubic cells: simple cubic has 1 atom/cell, body-centred cubic 2, face-centred cubic 4.
- Packing efficiency is the fraction of unit-cell volume occupied by spheres. Point defects include vacancy, interstitial, Schottky and Frenkel defects.
- Electrical behaviour includes conductors, insulators and semiconductors; doping changes conductivity.

Key formulae / facts

Density: $\rho = ZM/(a^3N_A)$

SC: Z=1, BCC: Z=2, FCC: Z=4

Packing efficiency: SC 52.4%, BCC 68%, FCC 74%

2. Solutions

- A solution is a homogeneous mixture. Concentration may be expressed as molarity, molality, mole fraction or mass percentage.
- Raoult's law relates partial vapour pressure to mole fraction for ideal solutions. Positive/negative deviations arise from unlike interactions.
- Colligative properties depend on the number of solute particles: relative lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure.
- Van't Hoff factor accounts for association or dissociation of solute particles.

Key formulae / facts

M = moles solute / L solution; m = moles solute / kg solvent

$\Delta T_b = iK_{bm}$; $\Delta T_f = iK_{fm}$

$\pi = iCRT$

$p = x\,p^\circ$; $\Delta p/p^\circ = x_{solute}$ (dilute)

3. Electrochemistry

- Electrochemical cells convert chemical energy into electrical energy; electrolytic cells use electrical energy to drive reactions.
- Oxidation occurs at the anode and reduction at the cathode. In a galvanic cell, anode is negative and cathode positive.
- Conductivity depends on ion concentration and mobility. Molar conductivity generally increases on dilution.
- The Nernst equation gives cell potential under non-standard conditions. Gibbs energy is related to cell emf.

Key formulae / facts

$E_{cell} = E_{cathode} - E_{anode}$

$\Delta G = -nFE_{cell}$

$E = E^\circ - (0.0591/n)\log Q$ at 298 K

$G = 1/\rho$; $\Lambda_m = \kappa \times 1000/C$ (C in mol L^{-1})

$Q = It$; $m = MIt/(nF)$

4. Chemical Kinetics

- Reaction rate is change in concentration per unit time. Rate law is determined experimentally, not generally from the balanced equation.
- Order is the sum of powers in the rate law; molecularity applies to an elementary step.
- For a first-order reaction, half-life is independent of initial concentration. Temperature increases rate constant, commonly described by Arrhenius equation.

Key formulae / facts

Rate = k[A]^m[B]^n; order = m+n

Zero order: [A]t = [A]₀ − kt

First order: k = (2.303/t)log([A]₀/[A]t)

t_½ (first order) = 0.693/k

k = Ae^(−E_a/RT)

5. Surface Chemistry

- Adsorption is accumulation of molecules at a surface; absorption occurs throughout the bulk.
- Physisorption involves weak intermolecular forces; chemisorption involves chemical bond formation.
- Colloids have dispersed particles intermediate in size between true solutions and suspensions. Tyndall effect, Brownian motion and electrophoresis are characteristic phenomena.
- Emulsions are colloidal dispersions of one liquid in another; emulsifying agents stabilise them.

Key formulae / facts

Freundlich isotherm: x/m = kP^(1/n)

Coagulation: precipitation of colloidal particles

Hardy–Schulze rule: higher counter-ion charge gives stronger coagulating power

6. General Principles and Processes of Isolation of Elements

- Metallurgy includes concentration of ore, conversion to oxide, reduction and refining.
- Concentration methods include hydraulic washing, magnetic separation, froth flotation and leaching.
- Calcination heats ore in limited/no air; roasting heats sulphide ore in excess air.
- Ellingham diagrams compare oxide stability and help predict suitable reducing agents. Refining methods include electrolytic, zone and vapour-phase refining.

Key formulae / facts

Roasting: sulphide ore + O_2 → oxide + SO_2

Calcination: carbonate ore → oxide + CO_2

Thermodynamic feasibility: $\Delta G^\circ < 0$

7. The p-Block Elements

- p-Block elements occupy groups 13–18. Their properties vary across periods and down groups due to size, ionisation enthalpy, electronegativity and inert-pair effect.
- Group 15: nitrogen family; common oxidation states −3, +3, +5. Group 16: oxygen family; common oxidation states −2, +4, +6.
- Group 17 elements are halogens and strong oxidising agents. Group 18 noble gases have filled valence shells and low reactivity.
- Study preparation, structures, oxidation states and important reactions of the compounds prescribed in the textbook.

Key formulae / facts

Inert-pair effect: reluctance of ns^2 electrons to participate in bonding

Oxidising power of halogens generally: $F_2 > Cl_2 > Br_2 > I_2$

8. The d- and f-Block Elements

- Transition elements have partially filled d orbitals in atoms or common ions. They show variable oxidation states, coloured ions, complex formation and catalytic activity.
- The lanthanoid contraction is the gradual decrease in size across the lanthanoid series due to poor shielding by 4f electrons.
- Actinoids show a wider range of oxidation states and many are radioactive.

Key formulae / facts

Magnetic moment (spin-only): $\mu = \sqrt{[n(n+2)]}$ BM

Unpaired d electrons often cause paramagnetism; d^0/d^{10} ions are usually colourless

9. Coordination Compounds

- A coordination entity contains a central metal atom/ion bonded to ligands. Coordination number counts donor atoms directly attached to the metal.
- Ligands may be monodentate, bidentate or polydentate. Naming follows ligand order and oxidation-state notation.
- Werner's theory distinguishes primary (ionisable) and secondary (coordination) valencies. Isomerism includes ionisation, linkage, geometrical and optical forms.
- Crystal field splitting helps explain colour and magnetic behaviour of complexes.

Key formulae / facts

Oxidation state: balance total ligand and complex charge

Magnetic moment: $\mu = \sqrt{[n(n+2)]}$ BM

10. Haloalkanes and Haloarenes

- Haloalkanes contain C–X bonds. Nucleophilic substitution occurs by SN1 (two-step, carbocation) or SN2 (one-step, backside attack).
- SN1 is favoured by stable carbocations; SN2 is favoured by less hindered substrates and strong nucleophiles.
- Haloarenes are less reactive towards nucleophilic substitution because the C–X bond has partial double-bond character.

Key formulae / facts

SN1 rate = k[RX]

SN2 rate = $k[RX][Nu^-]$

Aqueous KOH: substitution; alcoholic KOH: elimination (typical conditions)

11. Alcohols, Phenols and Ethers

- Alcohols contain –OH attached to saturated carbon; phenols have –OH directly attached to an aromatic ring; ethers contain R–O–R′.
- Alcohols undergo oxidation, dehydration and substitution. Phenols are more acidic than alcohols because phenoxide ion is resonance-stabilised.
- Ethers are commonly prepared by Williamson synthesis; cleavage with HI/HBr depends on structure and reaction conditions.

Key formulae / facts

Alcohol oxidation: 1° → aldehyde → acid; 2° → ketone

Williamson: RONa + R′X → ROR′ + NaX

Phenol + Br_2(aq) → 2,4,6-tribromophenol (white ppt)

12. Aldehydes, Ketones and Carboxylic Acids

- Aldehydes and ketones contain carbonyl groups. Nucleophilic addition occurs at the electrophilic carbonyl carbon.

- Aldehydes are generally more readily oxidised than ketones. Tollens' and Fehling's tests help distinguish many aldehydes.
- Carboxylic acids are acidic because the carboxylate ion is resonance-stabilised. They undergo esterification and reactions at the acyl group.

Key formulae / facts

Aldehyde + [O] → carboxylic acid

$RCOOH + R'OH \rightleftharpoons RCOOR' + H_2O$ (acid catalyst)

Aldol requires suitable α-hydrogen; Cannizzaro: aldehyde without α-H

13. Amines

- Amines are derivatives of ammonia and are classified as primary, secondary or tertiary.
- Basicity depends on availability of the nitrogen lone pair and solvation/inductive effects. Aromatic amines are generally less basic than comparable aliphatic amines.
- Diazonium salts are useful intermediates for preparing substituted aromatic compounds; they undergo coupling and replacement reactions.

Key formulae / facts

$RNH_2 + HCl \rightarrow RNH_3^+Cl^-$

Aniline + $NaNO_2$/HCl (0–5°C) → benzene diazonium chloride

14. Biomolecules

- Carbohydrates include monosaccharides, oligosaccharides and polysaccharides. Glucose is an aldohexose; fructose is a ketohexose.
- Proteins are polymers of α-amino acids joined by peptide bonds. Primary structure is amino-acid sequence; higher structures describe folding.
- Nucleic acids (DNA and RNA) are polymers of nucleotides. Enzymes are biological catalysts; vitamins support essential metabolic functions.

Key formulae / facts

Peptide bond: –CO–NH–

Nucleotide = sugar + nitrogenous base + phosphate

DNA: usually double-stranded; RNA: usually single-stranded

15. Polymers

- Polymers are macromolecules made from repeating monomer units. Addition polymerisation joins unsaturated monomers; condensation polymerisation eliminates small molecules.
- Classify polymers as natural/synthetic, linear/branched/cross-linked, thermoplastic/thermosetting and elastomers/fibres.
- Examples include polythene, PVC, Teflon, nylon-6,6, terylene, Buna-S and natural rubber.

Key formulae / facts

Addition: $n(CH_2=CH_2) \rightarrow (-CH_2-CH_2-)_n$

Condensation: monomers combine with loss of small molecules

16. Chemistry in Everyday Life

- Drugs interact with biological targets; therapeutic effect and side effects depend on dose and selectivity.
- Antacids neutralise excess stomach acid; antihistamines block histamine action; analgesics relieve pain; antibiotics act against bacteria.
- Soaps form micelles and work best in soft water; detergents generally work in hard water. Food preservatives slow spoilage.

Key formulae / facts

Soap: sodium/potassium salts of long-chain fatty acids

Micelle: hydrophobic tails inward, hydrophilic heads outward

Chemistry: reaction and numerical checklist

- Balance every chemical equation and write physical states when useful.
- Write reaction conditions above or below the arrow (temperature, catalyst, solvent or reagent).
- For electrochemistry, kinetics and solutions, define symbols and maintain consistent units.
- For organic conversions, identify the functional group first and choose a reagent that gives the required change.
- For coordination compounds, calculate oxidation state and coordination number before naming the complex.

Chemistry practice questions

- Differentiate crystalline and amorphous solids. Calculate the number of atoms in BCC and FCC unit cells.
- Explain Raoult's law and derive the relation for elevation in boiling point.
- Explain the working of a galvanic cell and derive the Nernst equation form used at 298 K.
- Differentiate order and molecularity. Obtain the first-order integrated rate equation.
- Compare physisorption and chemisorption; explain the Tyndall effect.
- Explain the role of an Ellingham diagram in metallurgy.
- Explain lanthanoid contraction and its consequences.
- Describe geometrical and optical isomerism in coordination compounds.
- Compare SN1 and SN2 mechanisms.
- Explain the tests/reactions used to distinguish aldehydes and ketones.

Practical preparation

Prepare the prescribed volumetric analysis, salt analysis, functional-group tests and project work. Record observations neatly, use clean apparatus, note units and follow laboratory safety instructions. Your school practical manual is the authority for the exact list.

PART III — MATHEMATICS

How to study: Read the concept points, learn the boxed relations, solve textbook examples, then attempt the practice questions at the end of this subject.

1. Relations and Functions

- A relation from A to B is a subset of A×B. A function assigns exactly one output in B to each input in A.
- Functions may be one-one, many-one, onto or into. A bijection is both one-one and onto.
- Composition is defined by $(g\circ f)(x)=g(f(x))$; inverse exists for a bijective function.
- Binary operations combine two elements of a set and may be tested for closure, commutativity, associativity and identity.

Key formulae / facts

$(g\circ f)(x) = g(f(x))$

f^{-1} exists ⇔ f is bijective

2. Inverse Trigonometric Functions

- Inverse trigonometric functions are defined using restricted domains so that each has a unique principal value.
- Memorise principal-value ranges and standard identities. Simplify expressions by converting inverse functions to angles in their principal ranges.
- Use identities carefully; arcsin(sin x) is not always x unless x lies in the principal range.

Key formulae / facts

$\sin^{-1}x + \cos^{-1}x = \pi/2$

$\tan^{-1}x + \cot^{-1}x = \pi/2$ (principal-value convention)

$\tan^{-1}x + \tan^{-1}y = \tan^{-1}[(x+y)/(1-xy)]$ with quadrant adjustment

3. Matrices

- A matrix is a rectangular array. Equality requires same order and equal corresponding entries.
- Addition requires same order. Matrix multiplication AB is defined when columns of A equal rows of B; generally AB ≠ BA.
- Transpose interchanges rows and columns. Symmetric matrices satisfy $A^T=A$; skew-symmetric matrices satisfy $A^T=-A$.
- For a square matrix, inverse exists only if determinant is non-zero.

Key formulae / facts

$(AB)^T = B^TA^T$

$A^{-1} = \text{adj}(A)/|A|$, when $|A| \neq 0$

$AA^{-1} = I$

4. Determinants

- A determinant is defined for a square matrix. Use expansion along a row/column and properties to simplify evaluation.
- Interchanging two rows changes sign; multiplying a row by k multiplies determinant by k; adding a multiple of one row to another leaves it unchanged.
- Cramer's rule solves a system when the determinant of coefficients is non-zero. Area of a triangle can be calculated using a 3×3 determinant.

Key formulae / facts

For 2×2: $|a\ b;\ c\ d| = ad-bc$

Area = ½|x₁(y₂−y₃)+x₂(y₃−y₁)+x₃(y₁−y₂)|

A(adj A) = |A|I

5. Continuity and Differentiability

- A function is continuous at x=a if lim(x→a)f(x)=f(a). Differentiability implies continuity, but continuity does not always imply differentiability.
- Derivative is the instantaneous rate of change and slope of tangent. Apply chain, product and quotient rules.
- Implicit and logarithmic differentiation are useful for complicated expressions. Derivatives of inverse trigonometric functions should be memorised.

Key formulae / facts

(uv)' = u'v + uv'; (u/v)' = (u'v−uv')/v²

Chain rule: dy/dx = (dy/du)(du/dx)

d(sin⁻¹x)/dx = 1/√(1−x²)

d(tan⁻¹x)/dx = 1/(1+x²)

6. Applications of Derivatives

- Derivatives measure rates of change and help find tangent/normal equations.
- A function is increasing where f′(x)>0 and decreasing where f′(x)<0. Critical points occur where f′=0 or is undefined.
- Use first- or second-derivative tests to classify local maxima and minima. Optimisation questions require defining the quantity and domain.

Key formulae / facts

Tangent at x=a: y−f(a)=f′(a)(x−a)

Normal slope = −1/f′(a)

If f′ changes + to −: local maximum; − to +: local minimum

7. Integrals

- Integration is the inverse process of differentiation. Indefinite integrals include a constant of integration.
- Use substitution, integration by parts and partial fractions. Definite integrals represent signed area and obey symmetry properties.
- Choose methods by identifying a composite function, product or rational expression.

Key formulae / facts

∫xⁿdx = xⁿ⁺¹/(n+1)+C (n≠−1)

∫eˣdx = eˣ+C; ∫dx/x = ln|x|+C

By parts: ∫u dv = uv−∫v du

∫ₐᵇf(x)dx = ∫ₐᵇf(a+b−x)dx

8. Applications of Integrals

- Definite integrals calculate area under a curve and between curves. Draw a rough graph and identify upper/lower boundaries.
- Area is non-negative; split intervals where curves cross or change order. Use symmetry where applicable.

Key formulae / facts

Area under y=f(x): A = ∫ₐᵇ f(x)dx when f(x)≥0

Area between curves: A = ∫ₐᵇ |f(x)−g(x)|dx

9. Differential Equations

- A differential equation relates a function to its derivatives. Order is the highest derivative order; degree is its power after removing radicals/fractions in derivatives.
- Formation may eliminate arbitrary constants. Solve variable-separable equations by separating x and y terms and integrating.
- For linear first-order equations, use an integrating factor.

Key formulae / facts

dy/dx = f(x)g(y): ∫dy/g(y) = ∫f(x)dx + C

dy/dx + Py = Q; I.F. = e^(∫Pdx)

Solution: y(I.F.) = ∫Q(I.F.)dx + C

10. Vector Algebra

- A vector has magnitude and direction. Components are resolved along coordinate axes; unit vectors i, j, k are mutually perpendicular.
- Dot product gives a scalar and is useful for angle/projection. Cross product gives a vector perpendicular to both and is useful for area.
- Two vectors are perpendicular when dot product is zero; parallel when cross product is zero (for non-zero vectors).

Key formulae / facts

a·b = |a||b|cosθ

|a×b| = |a||b|sinθ

Scalar projection of a on b = (a·b)/|b|

Area parallelogram = |a×b|

11. Three-Dimensional Geometry

- A line in 3D can be written in vector or symmetric form. Direction ratios are proportional to direction cosines.
- The angle between lines follows from their direction vectors. A plane is described by a normal vector and a point.
- Distance of a point from a plane is the absolute value of the plane expression divided by the normal magnitude.

Key formulae / facts

Line: r = a + λb

Plane: ax+by+cz+d=0

Distance = $|ax_1+by_1+cz_1+d|/\sqrt{(a^2+b^2+c^2)}$

cosθ = $(b_1 \cdot b_2)/(|b_1||b_2|)$

12. Linear Programming

- A linear programming problem optimises a linear objective function subject to linear constraints.
- Plot each inequality, identify the feasible region and its corner points. For a bounded feasible polygon, optimum occurs at a corner point.
- Check every corner point in the objective function; do not assume the largest coordinate gives the maximum.

Key formulae / facts

Objective: Z = ax + by

Constraints: linear inequalities; x≥0, y≥0

Evaluate Z at every feasible vertex

13. Probability

- Conditional probability is the probability of A given B. Independent events satisfy P(A∩B)=P(A)P(B).

- Bayes' theorem updates probabilities using observed evidence. A random variable assigns numerical values to outcomes.
- For a binomial distribution, there are n independent trials with constant success probability p.

Key formulae / facts

$P(A|B) = P(A \cap B)/P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Bayes: $P(A_i|B) = P(A_i)P(B|A_i)/\Sigma P(A_j)P(B|A_j)$

Binomial: $P(X=r) = {}^{n}C_{r} p^{r}(1-p)^{n-r}$

Mathematics: problem-solving checklist

- Write the domain and restrictions before simplifying functions or inverse-trigonometric expressions.
- In matrices and determinants, check dimensions and use properties to reduce lengthy calculations.
- In calculus, identify the method before starting: substitution, parts, partial fractions or a standard identity.
- In optimisation, find critical points and check endpoints or the full feasible region as required.
- In vectors and 3D geometry, write direction vectors/normals clearly and use dot or cross products appropriately.
- In probability, define events and state whether independence or conditional probability is being used.

Mathematics practice questions

- Determine whether a given function is one-one and onto; find its inverse if it exists.
- Find the inverse of a non-singular 3×3 matrix using adjoint method.
- Solve a system of three linear equations using determinants.
- Test continuity and differentiability of a piecewise-defined function at the joining point.
- Find intervals of increase/decrease and local extrema of a polynomial function.
- Evaluate an integral using integration by parts and another using partial fractions.
- Find the area enclosed between a parabola and a straight line.
- Solve a first-order linear differential equation using an integrating factor.
- Find the angle between two vectors and the area of the triangle they form.
- Find the equation of a plane through a point with a given normal vector.
- Solve a two-variable linear programming problem graphically.
- Apply Bayes' theorem to a problem involving two or more sources.

PART IV — BOARD EXAM PREPARATION

A practical 8-week revision plan

Week	Main target	Daily routine
1–2	Complete/relearn core concepts in all three subjects	2 study blocks for the weakest subject; 1 block each for the other two
3–4	Finish remaining chapters and textbook examples	Formula/reaction revision + written problem practice
5–6	Chapter tests and previous-year questions	Timed subject tests; analyse every mistake
7	Full-length mock papers	One paper at a time under exam conditions; review the same day
8	Final revision	Formula sheets, named reactions, definitions, diagrams and sleep routine

Answer-writing checklist

- Read the command word: define, derive, explain, compare, calculate or draw.
- Use headings and numbered steps for long answers.
- Draw clean, labelled diagrams in pencil where appropriate.
- Write formula first, substitute values clearly and show the final unit.
- Do not leave objective questions unanswered if there is no negative marking; follow the instructions printed on the actual paper.
- Reserve the final few minutes to check question numbers, units, signs and unanswered parts.

Quick revision tracker

Subject	Concepts	Textbook problems	Timed test	Revision 2
Physics	☐	☐	☐	☐
Chemistry	☐	☐	☐	☐
Mathematics	☐	☐	☐	☐

Reference and syllabus note

The chapter map in this guide is a broad Class 12 PCM revision framework based on commonly used NCERT/BSEB-aligned topics. The BSEB website is the final authority for official syllabus notices, model papers and updates. Check the session 2026–27 syllabus and the prescribed textbooks before treating any chapter or subtopic as examinable.

Official BSEB website: biharboardonline.com
Syllabus overview reference: Careers360 — Bihar Board Class 12 syllabus 2026–27.

End of study material • Keep practising consistently.